

Bexra-Inspired School Operating System

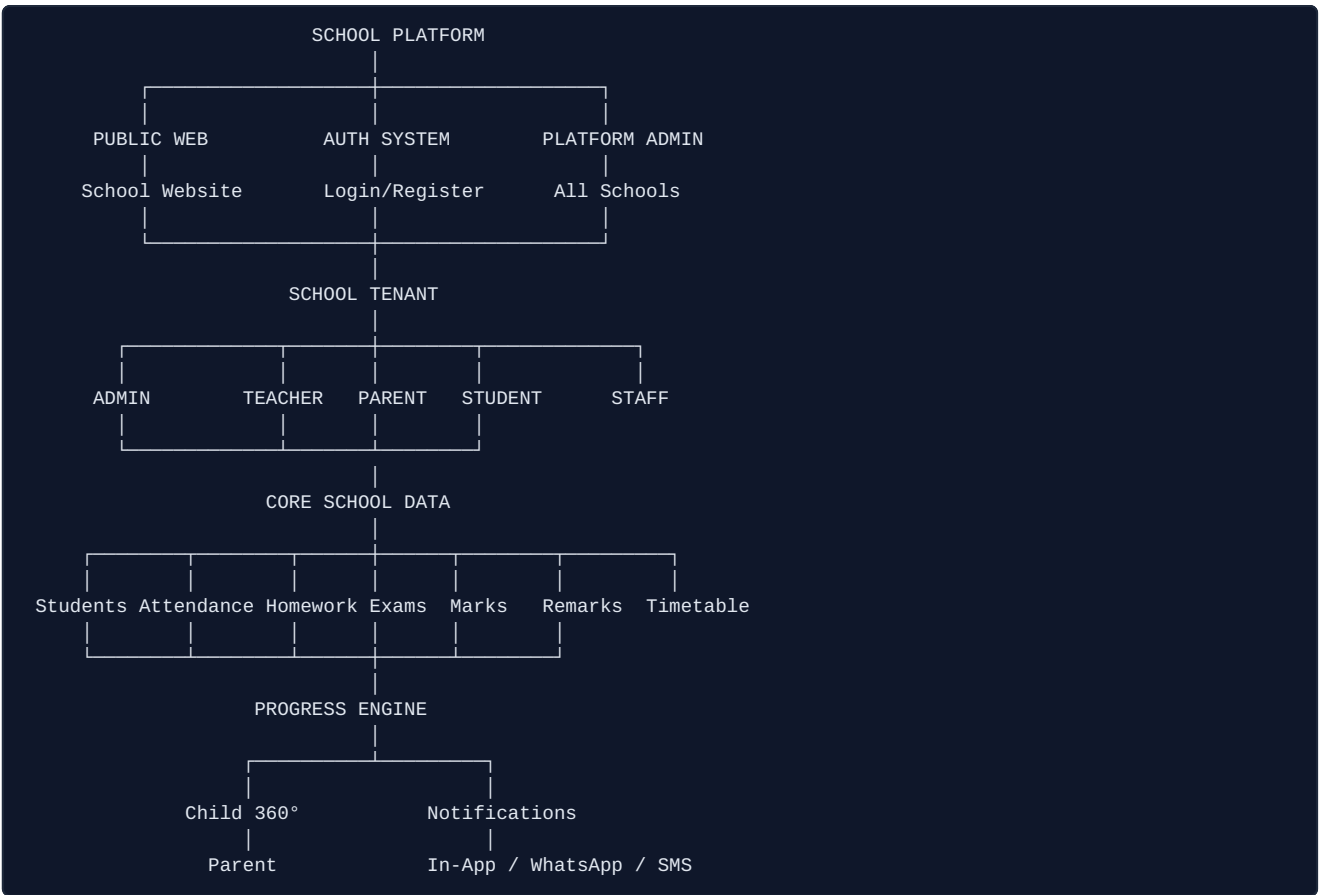
End-to-End Enterprise Architecture: Public CMS, Multi-Tenant Hierarchy, RBAC, Core School Workflows, Child 360° Progress Engine & Technical Specifications.

ARCHITECTURE	TARGET STACK	KEY DIFFERENTIATOR	DOC STATUS
Multi-Tenant SaaS	Spring Boot / PostgreSQL / Redis Parent Child 360° + Progress / React	Parent Child 360° + Progress Engine	Master Engineering Spec v1.0

PART 01

Overall Product Architecture

The platform is designed around a multi-tier SaaS model with strict data isolation per school tenant, federated public websites, unified authentication, and asynchronous progress intelligence.



PART 02

Public School Website (Bexra-Inspired)

Each school receives an auto-generated responsive public website rendered using tenant branding configurations.

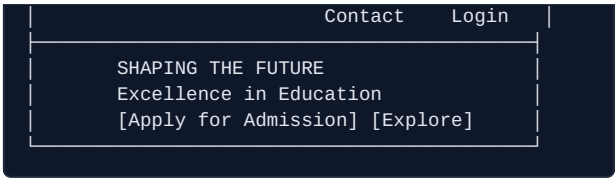


PART 03

Dynamic School Branding & Theming

School Admins configure visual identity under *Settings* → *Branding*. Assets are pulled on startup by the front-end router.

```
GET /api/public/schools/{slug}/config
```



Configured Pages: Home, About, Principal's Message, Academics, Classes, Faculty, Facilities, Gallery, Events, Announcements, Admissions, Contact, and Portal Login.

```
{
  "schoolName": "ABC Public School",
  "logo": "https://cdn.platform.com/schools/10/logo.png",
  "primaryColor": "#0284c7",
  "secondaryColor": "#0f172a",
  "favicon": "https://cdn.platform.com/schools/10/fav.ico",
  "theme": { "borderRadius": "8px", "fontFamily": "Inter" }
}
```

PARTS 04, 05 & 06

Multi-Tenancy Foundation, Authentication & Role Hierarchy

Multi-School Isolation (Tenant Guard)

Every operational table contains a mandatory indexed `school_id` foreign key. Tenant isolation is enforced at the database repository layer, never relying solely on frontend parameters.

```
User Token (schoolId = 10) → WHERE school_id = 10
```

Security guarantees zero cross-tenant data leakage across queries, batch jobs, and cached states.

Role Scope & Permissions Matrix

ROLE	ACCESS SCOPE	BOUNDARY
Super Admin	Platform-wide	All schools, subscriptions, revenue
School Admin	School-wide	Own school staff, students, configs
Teacher	Assigned Roster	Only assigned classes & subjects
Parent	Linked Wards	Only mapped children records
Student	Self Roster	Self attendance, work, report cards

Authentication Pipeline

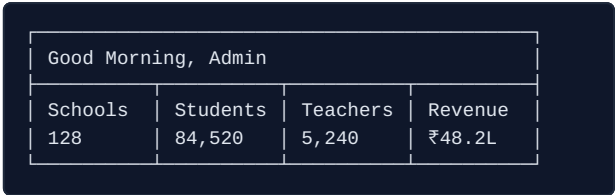
```
POST /api/auth/login → Credentials → User Lookup → Password Verification (BCrypt) → School Verification → Role Check → Signed JWT Token
```

```
JWT Payload: { "userId": 123, "schoolId": 10, "role": "PARENT", "linkedStudentIds": [501, 502], "exp": 1799980800 }
```

PARTS 07 & 08

Super Admin & School Onboarding

The Platform Master dashboard monitors subscription health, ARR, active tenant schools, and multi-tenant telemetry.



Onboarding Transaction: `POST /api/platform/schools`

PART 09

School Admin Operating Dashboard

High-level mission control displaying real-time metrics, live daily attendance percentages, and quick administrative actions.

```
Today's Attendance: Present 94.2% | Absent 5.8%
Fee Collection: ₹42.5L Collected | ₹12.3L Outstanding
```

Quick Operations: Add Student, Provision Teacher, Publish Emergency Announcement, Create Exam Schedule, Track Live Bus Fleet.

Create School → Create Academic Year → Generate Admin Account → Provision Default Roles & Permissions → Email Initial Setup Credentials

PARTS 10, 11, 12 & 13

Academic Structure, Roster & Multi-Ward Parent Mapping

Academic Schema & Teacher Assignments

Academics is partitioned by Academic Year, Classes (e.g. Class 8), and Sections (A, B, C). Teachers are granularly assigned to specific Subject-Section combinations.

```
teacher_assignments
-----
id | school_id | academic_year_id
teacher_id | subject_id | class_id | section_id
(Unique constraint on teacher_id + subject_id + section_id)
```

1-to-N Parent to Ward Mapping

A parent account links to multiple children across different classes, enabling seamless switching without logging out.

```
parent_students
-----
id | school_id
parent_id |
student_id | (Composite Key)
relationship [FATHER / MOTHER / GUARDIAN]
is_primary [BOOLEAN]
```

PARTS 14, 15, 16 & 17

Teacher Workspace, Attendance Engine & Real-Time Events

Teachers access a clutter-free interface optimized for 2-tap attendance and assignment management.

Daily & Period Attendance Flow

Supports both school-wide homeroom daily roll-call and individual period-by-period tracking.

```
POST /api/attendance
{ "classId": 8, "sectionId": 1, "date":
"2026-09-07", "records": [{"studentId": 101,
"status": "PRESENT"}, {"studentId": 102, "status":
"ABSENT"}] }
```

Period Attendance DB: Stores `period_number`, `subject_id`, and `teacher_id` to monitor subject-specific bunking or drop-offs.

Event-Driven Notification Fan-out

Attendance commits trigger asynchronous Spring events handled without blocking the teacher UI.

```
AttendanceMarkedEvent
├─ → Push Notification to Parent App (Instant)
├─ → WhatsApp Bot / SMS Service (If Absent)
└─ → Realtime Analytics Cache Update
```

PARTS 18, 19 & 20

Homework Lifecycle, S3 Storage & Status Pipeline

Status State Machine

```
[CREATED] → [PENDING] → [SUBMITTED] → [REVIEWED]
(If CurrentDate > DueDate and not Submitted → [OVERDUE])
```

Files uploaded by teachers (worksheets) and students (solutions) are uploaded directly to Object Storage (AWS S3 / GCS) via pre-signed URLs. Database retains only metadata and immutable S3 URI pointers.

Homework APIs & Lifecycle

ENDPOINT	ROLE	DESCRIPTION
<code>POST /api/homework</code>	Teacher	Publishes task + trigger notification
<code>GET /api/homework/student/{id}</code>	Parent/ Student	Fetches filtered assignments by state
	Student	Attaches submission document URL

```
POST /api/  
homework/{id}/  
submit
```

```
POST /api/  
homework/{id}/  
review
```

Teacher

Marks reviewed with
grades/comments

PARTS 21, 22, 23 & 24

Exams, Secure Marks Entry, Dynamic Report Cards & Behaviour Remarks

RBAC-Secured Marks Entry

Before persisting marks via `POST /api/exams/{id}/marks`, backend security verifies:

1. `user.school_id == exam.school_id`
2. Teacher holds active assignment for target class, section & subject
3. Exam marks entry window is currently open (not locked)

Constructive Qualitative Remarks

Replaces toxic numerical demerits with supportive qualitative rubrics across configurable dimensions:

- **Classroom Participation:** Excellent / Good / Developing
- **Teamwork & Leadership:** Highly Collaborative
- **Teacher Remarks:** Contextual personalized feedback

PARTS 25 & 26 — PLATFORM CROWN JEWEL

Parent Child 360° & The "Needs Attention" Heuristics Engine

The core innovation over legacy ERPs: Parents do not navigate 15 disparate menus. The single-page **Child 360°** computes an automated, holistic snapshot of academic and behavioural trajectory.

UI Architecture: Single Glance View

Today's Feed: Attendance Status (Present) • 5/6 Classes Attended • 2/3 Homework Completed • Latest Unit Test (18/20) • Teacher Remark ("Very engaged in science project")

Cumulative Index: Attendance: 94% | Homework: 91% | Academics: 84% | Conduct: Good

Algorithmic Rules Engine

⚠ ATTENTION TRIGGERS (Automated):

- Attendance drops below 85% in rolling 30 days
- Homework completion rate drops below 70%
- Continuous downward score gradient across 3 tests

✓ POSITIVE REINFORCEMENT (Encouraging tone):

- "Maths performance improved by 12% over last 3 assessments."

PARTS 27 TO 34

Auxiliary Modules: Communication, Timetable, PTM, Materials & Fees

MODULE	KEY SCHEMA / ENDPOINT	BUSINESS LOGIC & FLOW
Notifications	<code>/api/notifications</code>	Multi-channel broker: Push, WhatsApp, In-App. Deduped & logged with read receipts.
Announcements	<code>/api/announcements</code>	Targeted broadcasting: Entire School, Specific Grade, Section, or Role-only.
Timetable	<code>timetables</code>	Conflict-free scheduling engine for rooms, class sections, and teacher periods.
PTM Booking	<code>ptm_slots</code>	Self-service 15-minute slot reservation for parents + structured post-meeting notes.

PARTS 35 & 36

Master Database Schema & RESTful API Blueprint

Complete Relational Entities (PostgreSQL)

```
-- Core SaaS Entities
schools (id, slug, name, logo_url, config, is_active)
academic_years (id, school_id, label, start_date, end_date)
users (id, school_id, email, password_hash, status)
roles & user_roles (user_id, role_enum)

-- Stakeholder Profiles
teachers (id, user_id, employee_id, qualifications)
parents (id, user_id, phone, address)
students (id, user_id, admission_no, dob, roll_no)
parent_students (parent_id, student_id, relation)

-- Academic Roster
classes (id, school_id, academic_year_id, name)
sections (id, school_id, class_id, name)
subjects (id, school_id, name, code)
teacher_assignments (teacher_id, subject_id, section_id)
student_enrollments (student_id, section_id, academic_year_id)

-- Operations & Intelligence
attendance (id, school_id, student_id, date, status)
period_attendance (attendance_id, period_no, subject_id, status)
homework (id, school_id, section_id, subject_id, title, due_date)
homework_submissions (homework_id, student_id, file_url, status)
exams (id, school_id, name, term)
marks (exam_id, student_id, subject_id, score, max_score)
teacher_remarks (student_id, teacher_id, category, remark, date)
ptm_slots (id, teacher_id, start_time, is_booked, booked_by_parent)
fee_invoices & payments (invoice_no, student_id, amount, status)
audit_logs (id, school_id, user_id, action, entity, old_val, new_val)
```

Unified RESTful API Contract

```
# Authentication & Tenant
POST /api/auth/login
POST /api/auth/refresh
GET /api/public/schools/{slug}/config

# Platform & Tenant Administration
GET /api/platform/dashboard
POST /api/platform/schools
GET /api/schools/{id}/roster
POST /api/classes/{id}/sections

# Academics & Operations
GET /api/teacher/my-classes
POST /api/attendance
GET /api/attendance/report?classId=8&month=09
POST /api/homework
POST /api/homework/{id}/submission
POST /api/exams/{id}/marks

# Intelligence & Parent 360°
GET /api/parent/children
GET /api/parent/child-360/{studentId}
GET /api/analytics/needs-attention
POST /api/ptm/slots/book
GET /api/fees/student/{studentId}/invoices
```

PARTS 37, 38, 39 & 40

Backend Service Layer, Security Interceptors & Frontend Design System

Spring Boot Clean Architecture

```
Controller → Service (Business Rules) → Repository
→ PostgreSQL
Service → Spring ApplicationEventPublisher → Async
Listeners
```

Frontend Modular Architecture (React)

```
src/
├─ components/ui/      # Card, Button, Modal,
  Badge, Table
├─ modules/
|   └─ auth/           # Login, ForgotPassword
|   └─ platform/       # SuperAdmin onboarding &
```

Multi-Tenant Security Filter Flow

```
Request → JWT Filter (Validate Signature & Tenant ID)
  → SecurityContext populated with TenantContext
  → RBAC Interceptor (Role Matches Route)
  → Resource Ownership Check (e.g. Ward is mapped to Parent)
  → Execution Allowed / 403 Forbidden
```

```
analytics
|   ├── admin/           # Class, Staff, Fee, Exam
|   ├── teacher/        # 2-tap attendance, marks,
|   ├── parent/         # Child 360°, progress
|   ├── student/        # Homework upload,
|   ├── state/          # TanStack Query (React
|   ├── hooks/          # useTenant(),
|   └── state/           # useChild360(), useAuth()
```

PARTS 41 TO 49
UX Optimization, Analytics, Caching & Audit Safeguards

Targeted UI Paradigms

- **Teacher Mobile UI:** Designed for single-hand speed. No multi-level cascading menus. "Mark All Present" toggle allows complete class submission in < 5 seconds.
- **Parent Mobile UI:** Card-based 360° summary. Replaces overwhelming tabular data with graphical progress indicators and timely alerts.
- **Admin Desktop UI:** Dense, keyboard-navigable operational workspace with data tables, batch actions, and CSV/Excel exports.

Performance & Compliance

- **Redis Caching Layer:** School branding configs, active timetables, and teacher class rosters are cached with TTL + cache eviction on data mutators.
- **Immutable Audit Logging:** Mandatory recording of edits to grades, attendance modifications, and fee adjustments with old and new values.
- **Comparative Progress Engine:** Computes positive upward trends across terms to foster constructive student development.

Phased Implementation Roadmap & MVP Freeze

PHASE	DELIVERABLES	SCOPE & TECHNICAL FOCUS
Phase 1	Core Multi-Tenant Engine	Database migration, JWT auth, tenant isolation filters, RBAC definitions.
Phase 2	Academic Hierarchy & Profiles	Classes, sections, subjects, teacher assignments, student & parent onboarding.
Phase 3	Daily & Period Attendance	Teacher 2-tap attendance, absent alerts, rolling attendance calculations.
Phase 4	Homework & S3 Pipelines	Homework publishing, student file submission, review state machine.
Phase 5	Exams, Marks & Report Cards	Exam schedule, marks entry validation, automated aggregate calculations.
Phase 6 & 7	Child 360° & Needs Attention	Unified parent dashboard, trend comparison heuristics, behaviour logs.
Phase 8+	Auxiliary, Fees & V2 Modules	PTM booking, timetables, study materials, fee invoices, automated WhatsApp bots.

Strategic Differentiation: This architecture takes the best visual aesthetics, public website integration, and tenant branding from Bexra, and infuses it with a high-retention **Child 360° intelligence engine**. The resulting product elevates typical back-office school ERPs into an active engagement hub for school administrators, teachers, and parents.